missing. In 1984 this missing piece—string theory—dramatically entered the story and took center stage.

More Dimensions and String Theory

By now you should be convinced that our universe may have additional curled-up spatial dimensions; certainly, so long as they are small enough, nothing rules them out. But extra dimensions may strike you as an artifice. Our inability to probe distances smaller than a billionth of a billionth of a meter permits not only extra tiny dimensions but all manner of whimsical possibilities as well-even a microscopic civilization populated by even tinier green people. While the former certainly seems more rationally motivated than the latter, the act of postulating either of these experimentally untested-and, at present, untestable—possibilities might seem equally arbitrary.

Such was the case until string theory. Here is a theory that resolves the central dilemma confronting contemporary physics—the incompatibility between quantum mechanics and general relativity—and that unifies our understanding of all of nature's fundamental material constituents and forces. But to accomplish these feats, it turns out that string theory requires that the universe have extra space dimensions.

Here's why. One of the main insights of quantum mechanics is that our predictive power is fundamentally limited to asserting that such-and-such outcome will occur with such-and-such probability. Although Einstein felt that this was a distasteful feature of our modern understanding, and you may agree, it certainly appears to be fact. Let's accept it. Now, we all know that probabilities are always numbers between 0 and 1—equivalently, when expressed as percentages, probabilities are numbers between 0 and 100. Physicists have found that a key signal that a quantum-mechanical theory has gone haywire is that particular calculations yield "probabilities" that are not within this acceptable range. For instance, we mentioned earlier that a sign of the grinding incompatibility between general relativity and quantum mechanics in a point-particle framework is that calculations result in infinite probabilities. As we have discussed, string theory cures these infinities. But what we have not as yet mentioned is that a residual, somewhat more subtle problem still remains. In the early days of string theory physicists found that certain calculations yielded negative